LAB MANUAL

Curriculum Injection and Academic Integrity

Lab 3: Break the Assignment · Lab 4: Redesign and Guardrails

Date	Friday, October 30, 2026
Time	1:30 to 5:30 p.m. ET, online
Facilitator	Frazier A. Smith, Ed.D., Central Piedmont Community College
You need	A browser and a terminal tab on the range; the Participant Packet; your Redesign Workbook; one assignment of your own (or a fallback from the range)

Everything in these labs is synthetic. Northbridge Technical College, its instructors, and its students are fictional, and the four student submissions were written for this session. In the story they are education records, which is why every model run in these labs happens on the campus AI server. Do not paste real student work, names, or institutional data into any tool during this session.

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Before You Start

Range Access

1. Open the range URL your facilitator posts in chat and sign in with the account you used in Session 1.
2. Open a terminal tab and confirm the session files are present:

```
ls /opt/nitic/session2/
cd /opt/nitic/session2 && sha256sum -c SHA256SUMS | grep -vc OK
```

You should see `lab3/` and `lab4/`, and the second command should print `0` (no file failed its hash). Session 1's files are still in `/opt/nitic/session1/`.

3. Open Open WebUI in a second browser tab (link in chat) and sign in.
4. Open your copy of the Session 2 Redesign Workbook. The `Evidence Log` sheet has the same columns as Session 1; keep adding rows.

Model Names Used in This Manual

Placeholder	Range (Ollama) default	OpenRouter stand-in (if the range GPU is unavailable)
`SMALL_MODEL`	`gemma3:4b`	`openai/gpt-oss-20b`
`LARGE_MODEL`	`gpt-oss:20b`	`openai/gpt-oss-120b`

Set them once in the terminal:

```
export SMALL_MODEL=gemma3:4b
export LARGE_MODEL=gpt-oss:20b
```

> If the facilitator switches to OpenRouter, the student submissions would be leaving the range. In the Northbridge story that is a policy violation, so in that case the facilitator will run the model steps on screen and teams will work from the saved outputs. Note "OpenRouter (facilitator)" in the Environment column of any log row that depends on them.

Where Things Are

Path	What it is
`/opt/nitic/session2/lab3/assignment_CYB110_module6_incident_report.md`	The legacy assignment and its rubric
`/opt/nitic/session2/lab3/nbtc-fw01_excerpt.log`	The log the assignment is built on (the Session 1 item5 log)
`/opt/nitic/session2/lab3/submissions/`	Four student submissions, A to D
`/opt/nitic/session2/lab3/prompts/`	The prompts you will run: a `.txt` for pasting, and for most a `.md` explaining what to watch for
`/opt/nitic/session2/lab4/`	Syllabus excerpt, interim data policy, AIAS reference, redesign canvas, guardrails template, scenarios, evidence examples, fallback assignments

LAB 3 · 2:25 TO 3:10 · TEAMS OF THREE

Lab 3: Break the Assignment

Time: 45 minutes, teams of three **Deliverables:** Workbook sheet `Lab 3 Threat Model` completed; two AI outputs saved; four submissions scored on two scales; two Evidence Log entries; a one-sentence verdict

Scenario

Dana Okafor's CYB 110 Module 6 assignment asks students to review a firewall log excerpt and write a 400 to 600 word incident report. She believes at least fifteen of the thirty-four submissions were machine-written. The department chair has asked your team for evidence, before anyone argues about policy: can this assignment be completed by a model without the competency the course catalog promises, which is to "identify common attacks and recommend appropriate safeguards" through "log review"?

Team Roles

- **Runner** operates the models and saves outputs.
- **Checker** validates every claim in the outputs against the log with commands, and keeps a list of false claims.
- **Recorder** fills the `Lab 3 Threat Model` sheet and the Evidence Log.

Stage 1: Predict (8 minutes)

Read the assignment and its rubric:

```
cd /opt/nitic/session2/lab3
cat assignment_CYB110_module6_incident_report.md
```

In the workbook, for each of the four rubric rows, predict the score (5, 3, or 1) a large model will earn with a well-written prompt. Then write two lines: the competency the catalog promises, and the competency the rubric actually rewards. Do this before running anything; the sheet has a "predicted" column for a reason.

Stage 2: Run (15 minutes)

Run the lazy prompt (what a student in a hurry types) on the small model, and the CO-STAR solve prompt (what a student who took a prompting workshop writes) on the large model. Save both.

```
mkdir -p ~/lab3
cd /opt/nitic/session2/lab3

# Lazy prompt, small model
ollama run $SMALL_MODEL "$(cat prompts/lazy_prompt.txt nbtc-fw01_excerpt.log)" | tee ~/lab3/lazy_small.txt

# CO-STAR solve prompt, large model (prompt, then the assignment, then the log)
{ cat prompts/solve_costar.txt assignment_CYB110_module6_incident_report.md; echo; echo "=== LOG ==="; cat
nbtc-fw01_excerpt.log; } > ~/lab3/solve_input.txt
ollama run $LARGE_MODEL "$(cat ~/lab3/solve_input.txt)" | tee ~/lab3/solve_large.txt
```

While the large model generates (40 to 90 seconds), the Recorder scores the small model's output on the legacy rubric. Then score the large model's output. Record both scores next to your predictions.

Now the Checker validates. Every number and name in the outputs is a claim; the log is the only source. Run these against each output (shown for `solve_large.txt`; repeat for `lazy_small.txt`):

```
cd /opt/nitic/session2/lab3

# The facts
grep -c DROP nbtc-fw01_excerpt.log                              # 25
grep -c ACCEPT nbtc-fw01_excerpt.log                            # 25
grep -o 'SRC=[0-9.]*' nbtc-fw01_excerpt.log | sort | uniq -c | sort -rn
grep 'SRC=203.0.113.47' nbtc-fw01_excerpt.log | grep DROP | grep -o 'DPT=[0-9]*' | sort -u | wc -l   # 23
ports

# Every IP address the model mentioned, and how many times each appears in the log (0 = invented)
for ip in $(grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' ~/lab3/solve_large.txt | sort -u); do
  printf '%-16s %s\n' "$ip" "$(grep -c "$ip" nbtc-fw01_excerpt.log)"
done

# Claims the log cannot support: the firewall records connections, not login results
grep -niE 'fail|brute|unsuccessful|denied access|credential' ~/lab3/solve_large.txt

# Did the model find the finding that matters?
grep -c '10.30.12.118' ~/lab3/solve_large.txt
grep -c '1433' ~/lab3/solve_large.txt
```

For each false or unsupported claim, write one row in the `False claims` block of the sheet: the claim, the command that disproves it, and which output it came from. Typical findings: the scanner's attempts overcounted, "failed" VPN logins, an IP address with a count of 0, and no mention of the lab workstation reaching the SIS database port.

Stage 3: Read the Pile (15 minutes)

Four of the thirty-four submissions are on the range. Read them (they are also printed in the Participant Packet):

```
cd /opt/nitic/session2/lab3/submissions
cat submission_A_delgado.txt
cat submission_B_nair.txt
cat submission_C_ferreira.txt
cat submission_D_kowalski.txt
```

For each submission, record in the sheet the legacy rubric score (four rows, 5, 3, or 1 each), the answers to three competency questions, and your team's guess (AI, human, or hybrid). The three questions, each answered yes or no:

1. Did the author find and prioritize the lab workstation (10.30.12.118) reaching sis-db01 on port 1433?
2. Does the report avoid any fact that is not in the log (IPs, counts, reputations, login results)?
3. Does the report state what the log cannot show?

The Checker verifies at least one specific number in each submission with a command. The `for ip` loop above works on any file:

```
for ip in $(grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' submission_A_delgado.txt | sort -u); do
  printf '%-16s %s\n' "$ip" "$(grep -c "$ip" ../nbtc-fw01_excerpt.log)"
done
```

When the facilitator reveals the origins at 3:05, record how many of your four guesses were right. Do not revise the scores.

Stage 4: Verdict (7 minutes)

Complete the three lines at the bottom of the sheet:

Line	What goes there
The legacy assignment claims to measure	From the catalog
The legacy assignment actually measures	What the rubric rewards; what the AI outputs and the top-scoring submissions have in common
One sentence for the department chair	Your verdict, in words the chair can repeat to the advisory board

Add two Evidence Log rows (the small-model run and the large-model run), with the validation method (the commands) and a verdict of Reject with the reason. Highest data tier in input: Confidential (the log) for the runs; Restricted for anything you ran against a submission.

If your team finishes early: in Open WebUI, select `SMALL_MODEL`, paste `prompts/detector_prompt.txt` with submission C (the one your team thinks is most likely human), and run it three times in three fresh chats. Record the three first-line verdicts in the Evidence Log. Then try the paraphrase step in `prompts/detector_prompt.md` on submission A and run the detector once more.

Discussion Questions

1. Which rubric row was the best predictor of whether a submission was machine-written? Was any?
2. The human submission scored lowest on the legacy rubric. Which rubric words cost it points?
3. What would a student have to do, with this assignment unchanged, to earn a 20 without understanding the log?

LAB 4 · 4:05 TO 5:05 · TEAMS OF THREE

Lab 4: Redesign and Guardrails

Time: 60 minutes; teams of three for Stages 1 to 3, individual for Stage 4 **Deliverables:** Workbook sheets `Redesign Canvas` (Module 6), `Guardrails`, and `My Assignment`; a saved viva-question run; Evidence Log entries

Scenario

The chair wants three things before the advisory board meets on November 12: your Lab 3 evidence, a redesigned Module 6 that can run in Spring 2027 with the existing course VMs and adjunct staffing, and a department-wide syllabus AI section covering acceptable use, attribution, disclosure, prohibited data, evidence retention, accessibility, and academic integrity. The dean's constraint: nothing may conflict with the interim AI data policy adopted October 23, FERPA, or the college's ADA Title II obligations.

Read the two documents the redesign must respect:

```
cd /opt/nitic/session2/lab4
cat syllabus_CYB110_excerpt_current.md
cat northbridge_ai_data_policy_interim.md
```

Start Your Own Transcript

Lab 4 asks students to keep process evidence. Keep some yourself. Before Stage 1:

```
mkdir -p ~/lab4
script -a ~/lab4/lab4_transcript.txt
```

Everything you type in this terminal from now until you type `exit` is recorded with start and end timestamps. Look at the result at the end of the lab; it is the artifact Demo 3 showed, and you will have made one in under an hour of ordinary work.

Team Roles

- **Designer** leads the canvas (Stages 1 and 2).
- **Policy lead** leads the guardrails (Stage 3) and checks them against the data policy.
- **Runner and Recorder** runs the viva prompt, fills the workbook, and keeps the Evidence Log.

Stage 1: Lane and Performance (12 minutes)

Open the two references:

```
cat /opt/nitic/session2/lab4/aias_reference.md
cat /opt/nitic/session2/lab4/redesign_canvas.md
```

Decide the AI Assessment Scale level (1 to 5) and the Sydney lane (secured or open) for the redesigned Module 6, with a one-sentence reason, and enter them in the `Redesign Canvas` sheet. Then fill sections A to D.

Section	What to write
A. Competency	What the catalog promises; what Lab 3 showed the legacy version measures
B. Lane	The level, the lane, and the sentence students read on the assignment about what AI may and may not do
C. Authentic performance	The real artifact (what a shift analyst hands a lead), the audience and the constraint that make the decision matter, and what is unique per student so the task is not interchangeable. The last point is the one most teams skip; a seeded log that differs per student, a role, or a date all work
D. Decision-making	The judgment with two defensible answers, and how the rubric rewards the justification rather than the answer

Stage 2: Evidence, Validation, Accountability (15 minutes)

Fill sections E to H.

Section	What to write
E. Process evidence	What the student submits besides the product, how it is kept and for how long, and whether it is reviewed always, by sample, or on concern. The examples in `lab4/evidence_examples/` show the four cheapest kinds
F. Technical validation	The step where the student proves a claim or a fix works in the course VM, and what the graded output looks like
G. Accountability	The moment the student answers for the work in their own voice, and three questions only the author could answer well
H. Rubric rewrite	Four rows, 20 points, each measuring what the lane can claim

For section G, run the oral-defense prompt against one submission on the range. In the Northbridge story a submission is a Restricted record, so it runs only on the campus server, with history off: in Open WebUI, turn on Temporary Chat before pasting (the facilitator will show the toggle), or use the terminal, which saves nothing:

```
cd /opt/nitic/session2/lab3
{ cat prompts/viva_questions.txt; echo; echo "=== REPORT ==="; cat submissions/submission_D_kowalski.txt;
echo; echo "=== LOG ==="; cat nbtc-fw01_excerpt.log; } > ~/lab4/viva_input.txt
ollama run $LARGE_MODEL "$(cat ~/lab4/viva_input.txt)" | tee ~/lab4/viva_D.txt
```

Validate the five questions: each must point at a claim, number, or recommendation that is actually in submission D. Strike any that do not. Keep the best three for section G. Log the run (Restricted tier, local, validation method: checked each question against the report).

Stage 3: Guardrails (15 minutes)

```
cat /opt/nitic/session2/lab4/guardrails_template.md
cat /opt/nitic/session2/lab4/student_scenarios.md
```

Draft the seven areas in the `Guardrails` sheet, two to five sentences each, in language a student can act on. Then test: assign each of the eight scenarios to the area it exercises and decide whether your text gives the student and the instructor the same answer. Where it does not, revise. Two checks before you finish: nothing contradicts the

interim data policy (especially area 4, prohibited data, and area 7, what evidence the instructor relies on), and area 6 gives a student with approved assistive technology a path that does not require explaining a disability to the class.

Stage 4: The Sprint (15 minutes, individual)

Open the assignment you brought. If you did not bring one:

```
cat /opt/nitic/session2/lab4/own_assignment_fallbacks.md
```

In the `My Assignment` sheet, run canvas sections A to H on it. You will not finish every section in fifteen minutes; finish C (authentic performance) and E (process evidence) first, because they change the most. Section I, the cost check, is the honesty test: if the redesign doubles your grading time, decide what is sampled, moved to class time, or done by a local model you then verify.

Stage 5: Log and Close the Transcript (3 minutes)

Add Evidence Log rows for every model run in Lab 4. Then end your transcript:

```
exit
head -1 ~/lab4/lab4_transcript.txt
tail -1 ~/lab4/lab4_transcript.txt
```

Two timestamps and an hour of your own commands. Put the file name in the `Notes` column of your last log row.

Discussion Questions

1. Which of the five properties did your Module 6 redesign lean on most, and which did you leave weakest? Why?
2. Which guardrail area was hardest to write so that a student could follow it without guessing?
3. The cost check: what did the redesign cost in instructor minutes, and what did you cut to pay for it?

RUBRIC

Scoring

Lab 3 is scored on four points for the team's threat model:

Point	Earned when
1	Predictions were recorded before any model run
2	Every false claim in the two AI outputs is listed with the command that disproves it
3	All four submissions are scored on the legacy rubric and the three competency questions
4	The measures-versus-claims statement names the competency gap in one sentence

Lab 4 is scored on the 20-point rubric:

Criterion	Points	Full credit looks like
Threat model carried in	3	The redesign names the gap Lab 3 found and the AI behaviors it must defeat
Authentic performance and decision-making	4	A real artifact for a named audience under a stated constraint; something unique per student; a decision whose justification carries points
Process evidence and technical validation	4	Specific evidence named with retention terms; a validation step whose output is graded; no reliance on a detector
Accountability and rubric	3	An oral or live component with author-only questions; four rewritten rubric rows that measure what the lane can claim
Syllabus guardrails	4	All seven areas in student-actionable language; each tested against a scenario; consistent with the interim data policy, FERPA, and the Title II rule; assistive technology addressed
Own-assignment sprint and Evidence Log	2	Canvas sections A to H attempted for one real assignment, C and E complete; log rows complete with validation decisions

Export the three workbook sheets to the shared folder by 5:05.